Calibration of the dynamic analysis of Tellus 2D

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Abstract

This paper adds dynamic analysis to Tellus 2D, an agent-driven plane-strain geotechnical finite element program, and verifies and validates it before it is used in design. A dynamic phase integrates the equations of motion with the Newmark scheme from the stresses of any static phase, with consistent mass, Rayleigh damping per material and viscous, free-field, tied and compliant boundaries. The program reproduces exact solutions of wave propagation and damping within 3%, the surface motion of DEEPSOIL within 1.50%, the spectra of Strata for ten records with a mean log ratio of 0.008, the SHAKE91 sample problem within 2.03% and the period of a building in the PLAXIS 2D tutorial within 5%, although its static deflection under HS small is 27% larger than in the tutorial. At the Garner Valley downhole array, a one-dimensional analysis of 12 records over-predicts the surface spectra by a mean log ratio of 0.326 at 2% damping, which falls to 0.234 when the damping is matched at the site frequency. PM4Silt, implemented from the equations of its report, reaches the critical-state strength exactly and the cyclic strengths of the FLAC implementation within a factor of about 1.5 in the number of cycles. Applied to Austrian Dam in the 1989 Loma Prieta earthquake, it gives excess pore pressures of the measured magnitude and crest accelerations of 0.58–0.72 g, close to FLAC's 0.7 g, against 1.0–1.8 g for models without cyclic pore pressure, but the crest settlement could not be compared, because the loose unsaturated fill began to flow in strong shaking, which the implicit solution cannot follow. A 54 km section of the Bangkok basin, run through the 2025 M_w 7.7 Myanmar earthquake on one GPU in 90 minutes, reproduces the recorded velocity of about 10 cm/s and the amplification near 1 s, and behaves as one-dimensional columns within 10%.

1 Introduction

Dynamic finite element analysis is now a routine part of geotechnical design, because seismic codes ask for the response of slopes, embankments, walls and foundations to earthquakes, and because vibrations from machines, traffic and construction have to be predicted near sensitive structures. Commercial programs such as PLAXIS 2D and GTS NX provide this analysis through a time-domain module that shares the soil models and the staged construction of their static analysis, so that an engineer can shake the stresses that the construction left behind [4, 5]. A new program that offers the same analysis must show, in the same way as for its static analysis, that it solves the equations correctly and that its results agree with established programs and with measurements, because errors in a dynamic analysis, such as reflections from the model boundaries or a damping that depends on frequency, are much harder to see in the results than errors in a static one.

Tellus 2D is a plane-strain geotechnical program written in Rust whose primary user is a large language model (LLM) agent, which builds the model, calculates and reports through the tools of the Model Context Protocol (MCP) [2]. Its static analysis, with Mohr-Coulomb, Hardening Soil, HS small, Soft Soil and hypoplastic models, staged construction, consolidation and strength reduction, was calibrated in a companion paper against closed-form solutions, a commercial tutorial and four case histories [29]. This paper adds the dynamic analysis, describes its formulation and reports a calibration that has three parts, as for the static analysis. Exact solutions of wave propagation test the time integration, the mass, the damping and the boundaries, five other programs on fully specified problems test the analysis in the way it is used, and a vertical array of accelerometers tests it against the ground.

The site response programs SHAKE91 [18], Strata [20], DEEPSOIL [16] and Motus solve the one-dimensional problem of vertically propagating shear waves, either in the frequency domain or by a time-domain integration of a shear beam. A two-dimensional finite element program must reproduce their results when it is reduced to a soil column, because their solutions of that problem are well tested and in the linear case exact [22]. This reduction is a strong test of a two-dimensional code, because the column result does not depend on the uncertain inputs of a real site. The paper therefore relies on it heavily, and it adds the two-dimensional checks that a column cannot give, which are the absorption of waves at lateral boundaries, the transmission of the far field through free-field boundaries, a building on a deep foundation from a published tutorial and a sliding mechanism in an elastoplastic soil.

Section 2 summarises all benchmarks and their results in one table. Section 3 describes the formulation and how

it compares with commercial dynamic modules, Section 4 reports the verification against exact solutions, Section 5 the comparison with other programs, Section 6 the comparison with the downhole array, Section 7 the PM4Silt model of cyclic pore pressure, Section 8 the case of Austrian Dam, Section 9 a section of the Bangkok basin in the 2025 Myanmar earthquake, Section 10 the computing cost, and Section 11 discusses what the comparisons show and where their limits lie.

2 Summary of the benchmarks

Tellus 2D reproduces the exact solutions of wave propagation, damping and boundaries within 3%, except for the absorption of a surface wave at a viscous boundary, where the remaining reflection is 13.2–18.4% of the incident peak, and the elastoplastic sliding block slides 8.36% more than Newmark's rigid block, and it agrees with the linear site response programs within 1–10% in peak ground acceleration (PGA) and within 2–5% in mean spectral acceleration once the Rayleigh damping is placed at the site frequency (Table 1). The largest differences with other programs come from differences in their models rather than from the solution, namely the frequency dependence of Rayleigh damping against the frequency-independent damping of Strata, the elastic-perfectly plastic clay of Tellus 2D against the hyperbolic clay of Motus, and the static response of the HS small soil under the tutorial building. Against the downhole array the one-dimensional model over-predicts the short-period surface spectra, as one-dimensional analyses do at many vertical arrays.

Table 1: Benchmarks of the dynamic analysis and their results. Misfit: largest difference between two time histories relative to the peak of the reference. PSA: 5%-damped pseudo-spectral acceleration. Log ratio: mean of $\ln(\mathrm{PSA}_{\mathrm{Tellus}}/\mathrm{PSA}_{\mathrm{ref}})$ over 0.05–1 s (field) or 0.05–3 s (programs).

Benchmark	Reference	Result
Verification (Section 4)		
Layer on a rigid base, Ricker pulse	Exact reflection series	misfit 1.09%
Layer on a compliant base	Exact series	misfit 0.217% (same soil), 0.430% (rock)
Rayleigh-damped layer, harmonic	Exact damped transfer function	largest error 2.55% at 17 frequencies
Free-field sides, two layers	One-dimensional column	misfit <0.001%; viscous sides 92.2–116.1%
Line load on a half-space	Model of 150 by 100 m	misfit 13.2–18.4%; fixed sides 96.2–128.0%
Block on a Tresca layer	Newmark rigid block	slip 82.9 vs. 76.5 mm (+8.36%)
Other programs (Section 5)		
DEEPSOIL Example 2A	DEEPSOIL, linear	PGA 0.199 vs. 0.198 g; misfit 1.50%; PSA within 1.50%
Strata SCH site, 10 records	Strata, linear	PGA ratio 1.03 (largest error 9.09%); log ratio 0.008 ± 0.071; at f_0, $5f_0$: -0.039 ± 0.099
SHAKE91 sample problem	SHAKE91, strain-compatible	PGA 0.195 vs. 0.190 g; PSA mean error 2.03% (f_0, $5f_0$), 5.19% (0.5, 15 Hz)
PLAXIS lesson 17, free vibration	PLAXIS 2D	period 0.904 vs. 0.948 s; first swing 10.6 vs. 10.8 mm; deflection at release 7.31 vs. 5.75 mm (5.57 mm with elastic soil)
Bangkok soft clay, M7 at 150 km	Motus, nonlinear and EQL	PGA 0.260 vs. 0.107 (nonlinear) and 0.156 g (EQL); clay strain 0.809 vs. 7.94%
Field (Section 6)		
Garner Valley array, 12 records	Recorded surface motion	log ratio 0.326 ± 0.119 (ξ = 2% at 1, 15 Hz); 0.234 (2% at f_0, $5f_0$); 0.177 (4%)
PM4Silt element tests	FLAC cyclic strength curves [7, 9]	cycles to 3% mostly within a factor of 1.5 (at most 4.5) with the published parameters (Section 7)
Austrian Dam, Loma Prieta 1989	Piezometer heads, FLAC-PM4Silt analysis	crest acceleration 0.58–0.72 g with PM4Silt (FLAC 0.7 g), heads of the measured magnitude; settlement not compared (Section 8)
Bangkok basin section, Myanmar 2025	Bangkok records: PGA about 0.02 g, PGV about 10 cm/s	PGV 10–14 cm/s, PGA 1.3–1.7 times; 2-D within 10% of 1-D (Section 9)

3 Formulation

3.1 Equations of motion and time integration

A dynamic phase solves the equations of motion of the soil and the structures,

$$\boldsymbol{M}\ddot{\boldsymbol{u}}_{n+1} + \boldsymbol{C}\dot{\boldsymbol{u}}_{n+1} + \boldsymbol{F}_{\text{int}}(\boldsymbol{u}_{n+1}) = \boldsymbol{F}_{\text{s}} + \boldsymbol{F}_{\text{d}}(t_{n+1}), \tag{1}$$

in which $\boldsymbol{F}_{\text{int}}$ are the internal forces of the soil, interfaces, plates, piles, geogrids and anchors from the same stress integration as in static phases, $\boldsymbol{F}_{\text{s}}$ the static loads of the phase and $\boldsymbol{F}_{\text{d}}$ the dynamic forces. The phase starts from the displacements and stresses at the end of the phase it follows, at rest after a static phase and with the velocities and accelerations of a dynamic one, so a building can be constructed, loaded and then shaken in one calculation. The displacements are integrated with the Newmark scheme [24],

$$\boldsymbol{u}_{n+1} = \boldsymbol{u}_n + \Delta t\, \dot{\boldsymbol{u}}_n + \Delta t^2 \left[\left(\tfrac{1}{2} - \beta \right) \ddot{\boldsymbol{u}}_n + \beta\, \ddot{\boldsymbol{u}}_{n+1} \right], \tag{2}$$

$$\dot{\boldsymbol{u}}_{n+1} = \dot{\boldsymbol{u}}_n + \Delta t \left[(1-\gamma)\ddot{\boldsymbol{u}}_n + \gamma\, \ddot{\boldsymbol{u}}_{n+1} \right], \tag{3}$$

with the average acceleration rule $\beta = 0.25$ and $\gamma = 0.5$ by default, which is unconditionally stable and has no numerical damping, and with $\gamma > 0.5$ when the highest frequencies are to be damped. Each time step is solved for the displacement increment by the initial stiffness method of the static phases, with the effective matrix $\boldsymbol{K} + a_0\boldsymbol{M} + a_1\boldsymbol{C}$, $a_0 = 1/(\beta\Delta t^2)$ and $a_1 = \gamma/(\beta\Delta t)$, which is factorised once for a time step and kept while the stiffness of stress-dependent soils changes by less than 30%.

An elastoplastic step converges slowly with the initial stiffness matrix where an element has yielded at all its stress points, because the internal nodes of such an element have little mass and no plastic stiffness, so that the iteration removes only a few per cent of the error in that mode per iteration. A dynamic phase therefore accelerates the iterations with the Anderson scheme [1, 28], which combines the last six iterates so as to minimise the preconditioned residual and acts like a Krylov method on the slow modes, and it accepts the step under the rule of the static phases, which allow a few nodes ($3 + n_p/10$ for n_p plastic points) to stay out of balance after the last iteration when the rest of the mesh is in equilibrium. A step that still fails is repeated with half the time step, down to 1/64 of it, and the time step returns to its value after eight easy steps. Unless it is given, the time step follows from the signals, as their sampling interval or a twentieth of the period of their highest frequency, and it is at most a hundredth of the phase.

3.2 Mass and damping

The mass matrix is the consistent mass of the soil with the density $\rho = \gamma/g$, where γ is the saturated unit weight below the water table and the unsaturated one above it, so the mass of the pore water moves with the soil, and of the plates with their weight w/g per unit length (their rotational inertia is neglected). Each soil and plate material has Rayleigh damping, $\boldsymbol{C} = \alpha\boldsymbol{M} + \beta\boldsymbol{K}$, where $\boldsymbol{K}$ is the elastic stiffness at the start of the phase without the water of undrained soils, and the coefficients follow from two target damping ratios ξ_1, ξ_2 at two frequencies f_1, f_2, because the damping ratio at a circular frequency ω is $\xi(\omega) = \alpha/(2\omega) + \beta\omega/2$. Rayleigh damping is therefore exact only at the two frequencies, smaller between them and larger outside them [17], and Section 5 shows what this costs against programs with frequency-independent damping. The HS small model adds hysteretic damping through its unloading and reloading at the small-strain stiffness G_0 [3, 6].

3.3 Boundaries and input

The model sides and base have their static supports in a dynamic phase unless they are given one of four dynamic conditions, in which case the support is released and its reaction at the end of the previous phase is kept as a load. A *viscous* side carries dashpots normal and tangential to it, $C_1\rho V_p$ and $C_2\rho V_s$ per unit area [23], which absorb waves arriving at it at normal incidence and most of the energy of oblique waves, with the relaxation coefficients $C_1 = C_2 = 1$ by default. The wave speeds come from the shear modulus and the constrained modulus of the adjacent element at its current stiffness, including the stiffness of the pore water in undrained soil for V_p and the small-strain stiffness of HS small. A *compliant* base is a viscous base through which the earthquake enters as the outcrop motion: the base receives the stress $\rho V_s(\dot{u}_{\text{o}} - \dot{u})$, where $\dot{u}_{\text{o}}$ is the velocity of the outcrop motion, twice that of the incident wave, and $\dot{u}$ the velocity of the base [19]. The impedance is that of the soil at the base or of a given half-space below it. A *rigid* base keeps its support and moves with the prescribed within motion, which is integrated with the same Newmark scheme so that its displacements, velocities and accelerations are consistent with the motion of the soil.

A viscous side does not suit an earthquake from the base, because its dashpots resist the motion of the whole column (Section 4.2). A *free-field* side therefore couples the model to a one-dimensional column of the soil at that

side, which is solved at each time step with the same base condition and input, and which applies its stresses and the force of the dashpots driven by its own velocity to the side nodes of the model [30]. The coupling is one way, from the column to the model, and the column is linear elastic with the Rayleigh damping of its soils. A *tied* pair of sides makes the nodes of the two sides at the same level share their displacements, which models a horizontally layered site exactly with a narrow model, and which needs nodes at the same levels on both sides, as a structured mesh gives.

Dynamic loads are components of the loads of the model, an amplitude in x and y times a signal, which is harmonic, a Ricker pulse or a table of values in time. Earthquake records are read from PEER files or from tables, in g or in m/s^2, and a drift correction removes the linear trend $c_0 + c_1 t$ for which the velocity and displacement integrated from the record are zero at its end. Each dynamic phase records the displacement, velocity and acceleration of chosen points at every step, the peak velocity and acceleration of every node, and the full displacement field at up to a hundred steps.

3.4 Comparison with commercial programs and use by an agent

The dynamic analysis covers the plane-strain features of the dynamic module of PLAXIS 2D, which are the Newmark integration, Rayleigh damping per material, viscous, free-field, compliant base and tied boundaries, dynamic multipliers for loads and prescribed motions, drift correction, dynamic phases after static ones and the use of all soil models in them [4]. It does not include the dynamic consolidation of PLAXIS 2D, its liquefaction models (UBC3D-PLM and PM4Sand), axisymmetric models, or the eigenvalue and response spectrum analyses that GTS NX adds to its time history analysis. The MCP server of the program has three new tools, `add_signal`, `set_monitors` and `get_dynamic`, and the phase tools take a calculation type `dynamic` with its settings, so an agent can define and run an earthquake analysis and read its time histories, peak accelerations and the largest acceleration in the model. The project file and the desktop application hold the same settings.

4 Verification against exact solutions

4.1 Wave propagation in a soil column

A uniform elastic layer of thickness $H = 20$ m, unit weight 20 kN/m^3 and $V_s = 100$ m/s, modelled as a column 1 m wide with tied sides and a structured mesh of 1 m 25-node elements, was shaken at its base by a Ricker pulse of 0.1 g with a central frequency of 4 Hz. On a rigid base the surface acceleration is the series $a_s(t) = 2\sum_n (-1)^n a_g(t - (2n+1)\tau)$ with $\tau = H/V_s$, which is the time-domain form of the transfer function $1/\cos(\omega\tau)$ [21]. On a compliant base over a half-space of impedance ratio $r = \rho_1 V_1/(\rho_2 V_2)$ with an outcrop motion a_o, it is $a_s(t) = T\sum_n q^n a_o(t - (2n+1)\tau)$ with $T = 2/(1+r)$ and $q = (r-1)/(r+1)$, so that a layer on a half-space of its own soil passes the outcrop motion on with the delay τ and without reflections. Tellus 2D reproduces both series (Figure 1), with a largest difference of 1.09% of the peak on the rigid base, 0.217% on a half-space of the same soil, and 0.430% on rock of $V_s = 400$ m/s and 22 kN/m^3. The compliant base therefore absorbs the downgoing waves and transmits the input with the correct impedance, and the peak of the rigid base result, 1.96 m/s^2, equals the exact 1.96 m/s^2.

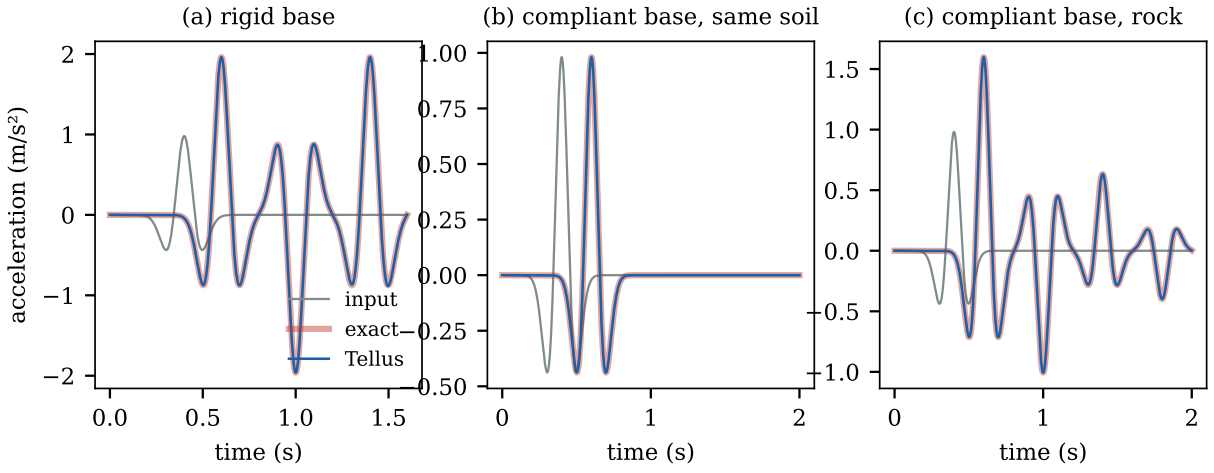


Figure 1: Surface acceleration of a 20 m layer ($V_s = 100$ m/s) under a 4 Hz Ricker pulse: (a) rigid base with the within motion, (b) compliant base over the same soil, (c) compliant base over rock of 400 m/s, against the exact series of reflections.

The same column with Rayleigh damping of 5% at 1 and 10 Hz ($\alpha = 0.571$ s^{-1}, $\beta = 1.45 \times 10^{-3}$ s) was shaken harmonically on a rigid base at 17 frequencies between 0.5 and 7 Hz for 14 s, after which the transients had decayed. A continuum with Rayleigh damping obeys $\rho(\ddot{u}+\alpha\dot{u}) = G(1+\beta\,\partial_t)u_{,zz}$, so its wavenumber is $k^2 = \rho(\omega^2 - i\omega\alpha)/[G(1+i\omega\beta)]$ and its amplification is $1/|\cos kH|$. The steady-state amplification of Tellus 2D agrees with this solution within 2.55% at every frequency, including the first three resonances at 1.25, 3.75 and 6.25 Hz where it reaches 15.0, 7.3 and 3.5 (Figure 2), so both parts of the damping matrix act as intended.

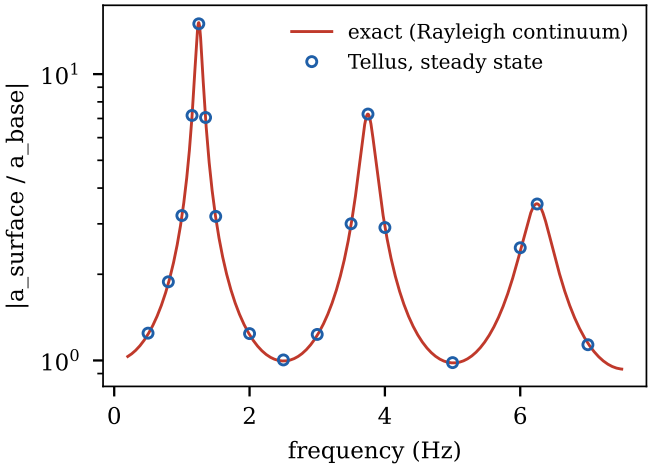


Figure 2: Steady-state amplification of a 20 m layer with Rayleigh damping (5% at 1 and 10 Hz) on a rigid base: Tellus 2D at 17 frequencies and the exact solution for a Rayleigh-damped continuum.

4.2 Free-field and viscous sides

A model 60 m wide and 20 m deep with two layers (8 m of V_s = 80 m/s over 12 m of 150 m/s) and an unstructured mesh of 2 m elements was shaken from a rigid base by a 3 Hz Ricker pulse, and its surface motion was compared with that of the same profile as a tied column. With free-field sides the surface motion at the side, 5 m from it, 15 m from it and at the centre differs from the column result by less than 0.001% of the peak, so the free-field columns carry the far field into the model without disturbing it (Figure 3). With viscous sides the misfit is 92.2% at the side and still 116.1% at the centre, because the dashpots resist the motion of the whole model, which moves with the base. This is why commercial manuals ask for free-field boundaries in earthquake analyses, and the result shows that the warning applies as much to the centre of a model three times as wide as it is deep as to its sides.

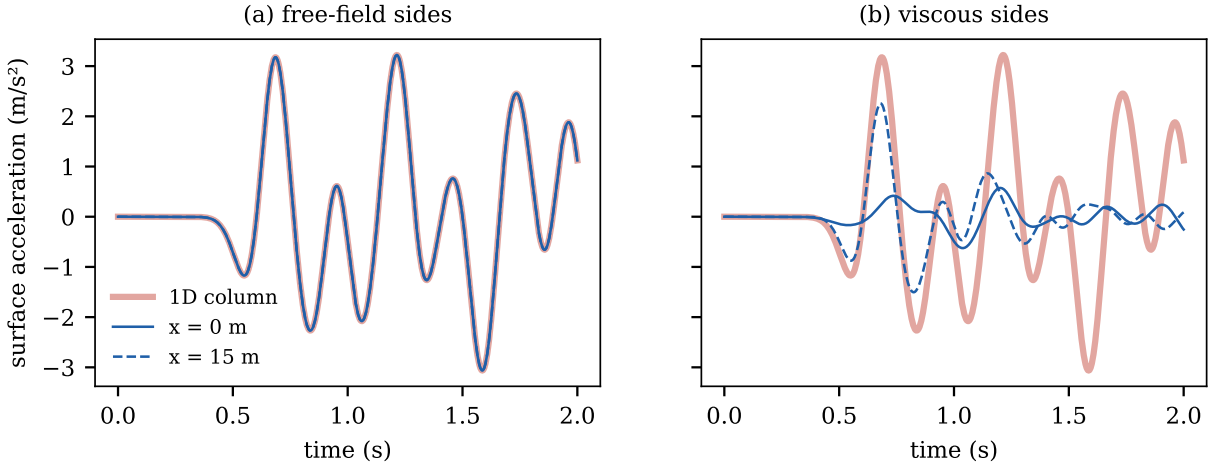


Figure 3: Surface acceleration of a two-layer model 60 m wide shaken from a rigid base, at the side ($x = 0$) and 15 m from it, with (a) free-field and (b) viscous sides, against the one-dimensional column.

4.3 Waves from a surface load

A vertical line load of 100 kN/m per metre on a strip 0.5 m wide, varying as a 10 Hz Ricker pulse, was applied at the surface of an elastic half-space (V_s = 200 m/s, ν = 0.25) modelled as a half on its axis of symmetry. A model 30 m wide and 20 m deep with viscous sides and base was compared with a model 150 m wide and 100 m deep, whose boundaries the waves do not reach in the 0.45 s of the analysis, and with the small model with fixed boundaries. The viscous boundaries leave reflections of 13.2, 18.4 and 6.27% of the peak vertical velocity at 5, 10 and 20 m from the load, against 96.2–128.0% with fixed boundaries (Figure 4). The remaining reflection comes from the Rayleigh wave and the oblique body waves, which a dashpot tuned to normal incidence cannot absorb fully [23], and it is the reason to keep the boundaries of a vibration analysis well away from the region of interest.

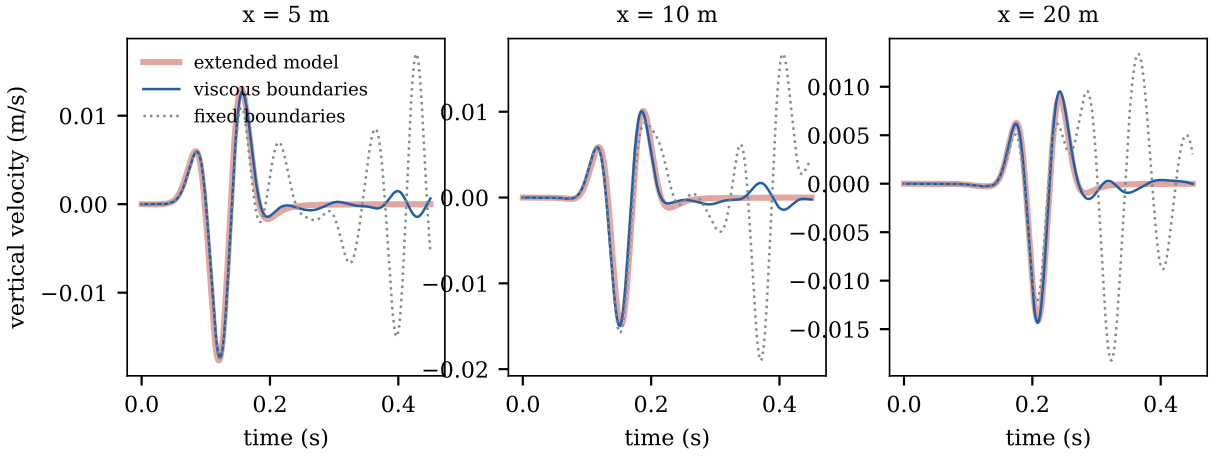


Figure 4: Vertical surface velocity at 5, 10 and 20 m from a Ricker line load on an elastic half-space: a 30 by 20 m model with viscous or fixed boundaries against a 150 by 100 m model whose boundaries the waves do not reach.

4.4 Sliding of a block on a weak layer

The dynamic analysis with an elastoplastic soil was checked against the rigid sliding block of Newmark [25]. A stiff elastic block 10 m thick (V_s = 800 m/s) rests on a layer 0.5 m thick of undrained soil with s_u = 40 kPa on a rigid base, with tied sides, and the base was shaken by two half-sine pulses of 0.4 g and 0.25 s. The layer limits the shear stress at its base to s_u, so the block cannot accelerate faster than $a_y = s_u/(\rho h)$ = 0.190 g for the 10.5 m of soil above, and it slides one way whenever the base accelerates faster. Tellus 2D gives a slip of 82.9 mm at the end against 76.5 mm from Newmark's integration (Figure 5), 8.36% more, and the 95th percentile of the acceleration of the top of the block is 0.226 g against a_y = 0.190 g. The difference is expected, because the block is elastic and yields over a layer of finite thickness rather than at a plane, and because its first mode at 20 Hz rings after each start and stop of sliding, which lifts the peak acceleration of the top to 0.376 g.

5 Comparison with other programs

5.1 Linear site response: DEEPSOIL, Strata and SHAKE91

The site response programs were compared on three published problems whose inputs and results are available as numbers [16, 18, 20], with each profile modelled as a tied column of 1 m elements (0.75 m for SHAKE91) on a compliant base over the given half-space. Example 2A of DEEPSOIL is a 20 m layer with V_s = 500 m/s and no damping on a half-space of 760 m/s, shaken by a Chi-Chi record of 0.183 g as an outcrop motion. Tellus 2D gives a surface PGA of 0.199 g against 0.198 g from DEEPSOIL, its surface acceleration history differs from that of DEEPSOIL by at most 1.50% of the peak, and its response spectrum agrees within 1.50% at all periods, with a peak of 0.571 g against 0.569 g at 0.534 s (Figure 6). The 2% damping of the half-space in DEEPSOIL is not modelled, and it explains the small differences at the resonances.

The SCH site of the Strata examples has four layers to 91 m (V_s = 200, 300, 460 and 700 m/s) with 2% damping over a half-space of 760 m/s, and it was shaken by the ten records of the Strata example, which include Chi-Chi, Irpinia, Loma Prieta, Morgan Hill and Northridge motions with PGA of 0.09 to 0.36 g. Strata damps every frequency

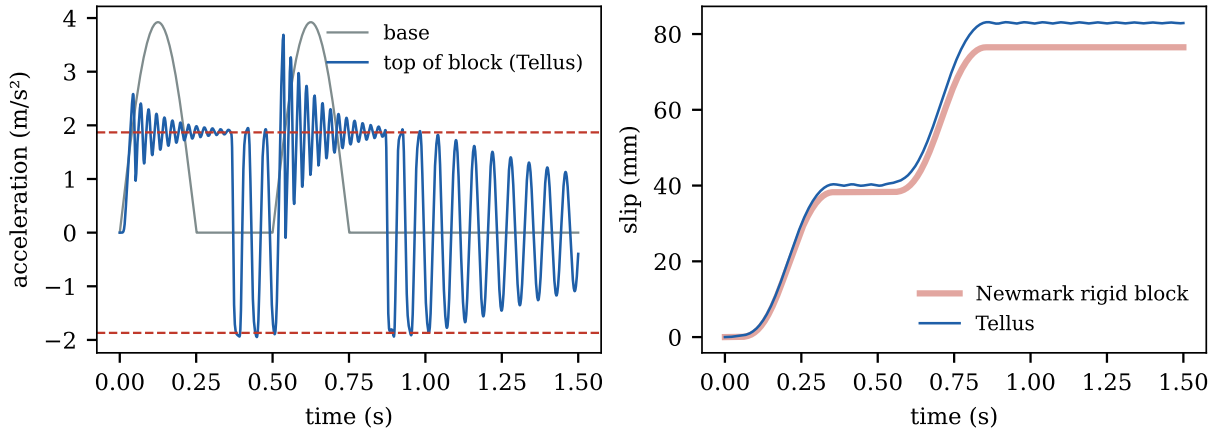


Figure 5: A 10 m block on a 0.5 m Tresca layer (s_u = 40 kPa) under two pulses of 0.4 g: (a) acceleration of the base and of the top of the block with the yield acceleration $\pm a_y$, (b) slip of Tellus 2D and of Newmark's rigid block.

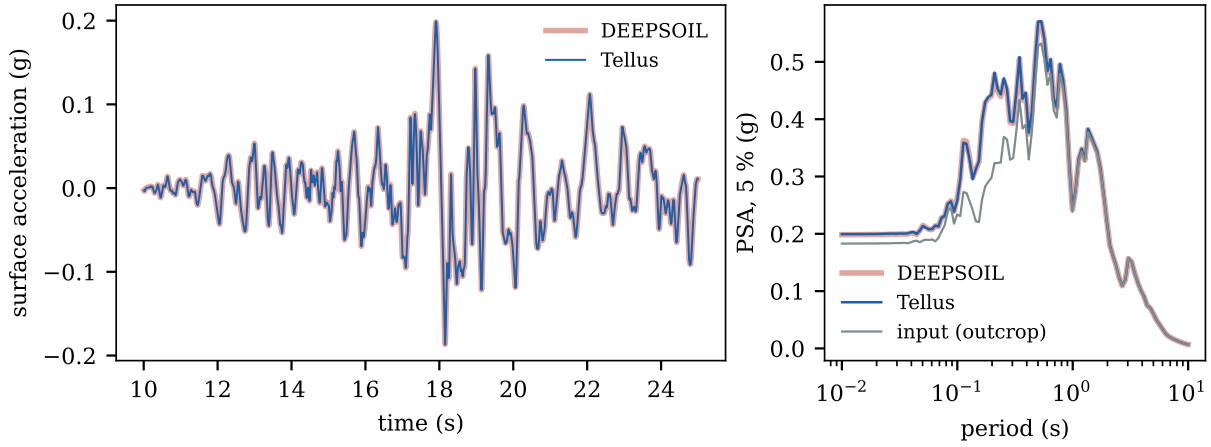


Figure 6: DEEPSOIL Example 2A: (a) surface acceleration between 10 and 25 s, (b) 5%-damped surface response spectrum of Tellus 2D and DEEPSOIL, and of the outcrop input.

by 2%, whereas Tellus 2D used Rayleigh damping of 2% at 0.5 and 10 Hz, which damps less between these frequencies and more outside them. Over 0.05–3 s the mean log ratio of the Tellus 2D spectra to those of Strata is 0.008 with a standard deviation of 0.071 over all records and periods, and the surface PGA of Tellus 2D is on average 1.03 times that of Strata, with a largest difference of 9.09% (Figure 7). The ratio departs from zero where the Rayleigh damping differs from 2%, below it between its two frequencies and above it outside them, which is the known error of this damping and not of the solution [17, 22]. With the damping matched instead at the fundamental frequency of the site, $f_0 = V_s/(4H)$ = 1.13 Hz from the travel time, and at $5f_0$, a common choice in practice, the mean log ratio is -0.039 ± 0.099 and the PGA ratio 0.967, so for this site both choices are within about 4% of Strata on average.

The sample problem of SHAKE91 is a 45.7 m profile of 16 clay and sand layers over rock of 1,219 m/s, shaken by the Diamond Heights record of the 1989 Loma Prieta earthquake scaled to 0.1 g, which SHAKE91 solves by equivalent linear iterations. Tellus 2D used the strain-compatible shear moduli and damping ratios of the last SHAKE91 iteration as linear properties, with Rayleigh damping matched to each layer's ratio at 0.5 and 15 Hz, so the comparison tests the propagation through the converged profile. With the Rayleigh damping of each layer matched at the site frequency f_0 = 1.85 Hz and at $5f_0$, it gives a surface PGA of 0.195 g against 0.190 g from SHAKE91 and a mean difference in spectral acceleration of 2.03% over 0.05–3 s, and with the damping matched at 0.5 and 15 Hz the difference is 5.19% (Figure 8). These differences are against the spectrum of the surface acceleration history that SHAKE91 writes, computed with the same routine as for Tellus 2D. The spectrum that SHAKE91 prints (its Option 9) differs from that history by 25.0% on average, almost all of it at periods of 0.07–0.15 s, which are only 3.5 to 7.5 times the 0.02 s time step of the record. An exact linear solution in the frequency domain with the strain-compatible properties of SHAKE91 reproduces the history and not the printed spectrum, so the printed spectrum, and not the propagation, is the outlier, and comparisons with this sample problem should use its time history.

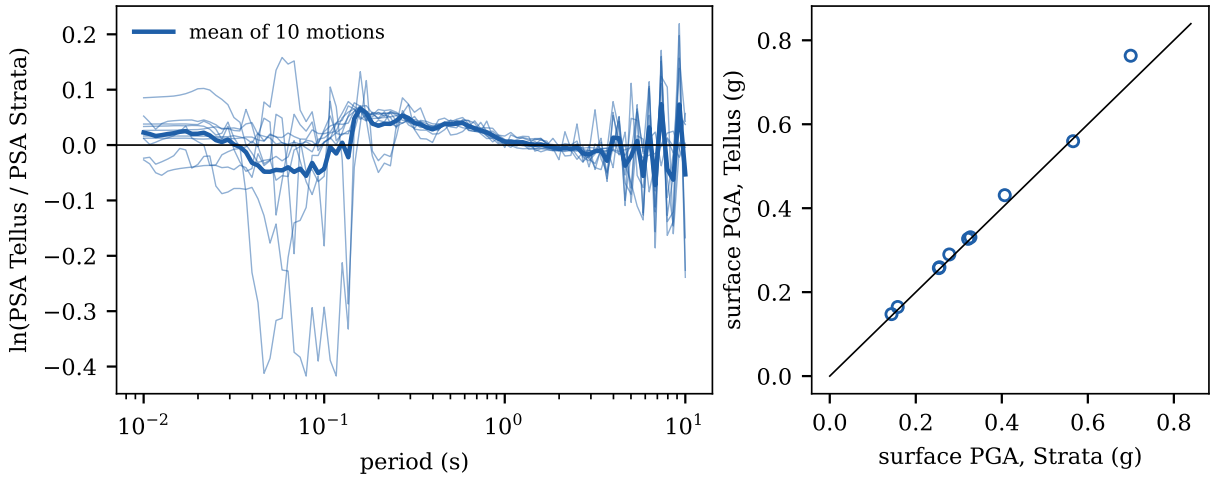


Figure 7: Strata SCH site with ten records: (a) log ratio of the surface spectra of Tellus 2D (Rayleigh 2% at 0.5 and 10 Hz) to those of Strata (2% at all frequencies), each record and their mean, (b) surface PGA.

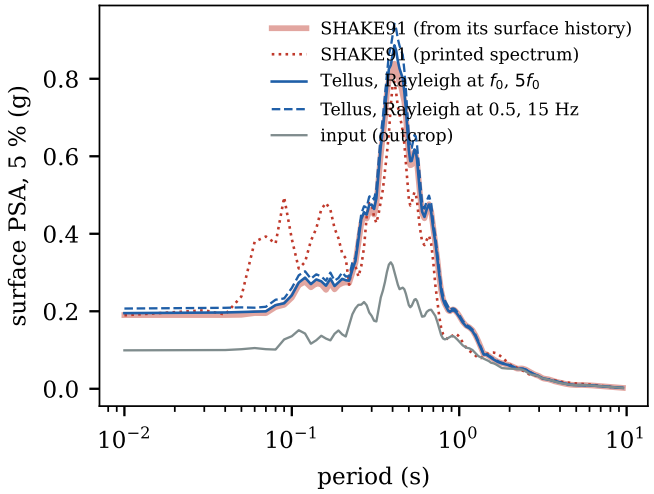


Figure 8: SHAKE91 sample problem: 5%-damped surface spectra of SHAKE91, from its surface acceleration history and as printed, and of Tellus 2D with the strain-compatible properties of SHAKE91 as linear properties and its Rayleigh damping matched at f_0 and $5f_0$ or at 0.5 and 15 Hz.

5.2 Free vibration of a building: PLAXIS tutorial lesson 17

Lesson 17 of the PLAXIS 2D tutorial is a five-storey building, 10 m wide and 15 m high with a 2 m basement, on 15 m of loose sand over 25 m of medium dense sand, both modelled with HS small, in a model 160 m wide and 55 m high with viscous sides and base [5]. The building is made of plates with Rayleigh damping ($\alpha = 0.232$, $\beta = 0.008$) and a central column of node-to-node anchors, it is constructed in phase 1, pushed by a horizontal load of 10 kN/m at its top-left corner in phase 2 and released in phase 3, a dynamic phase of 5 s in which it vibrates freely. Tellus 2D used the inputs of the manual, a mesh of 2,070 elements with a coarseness of 0.3 in the soil, a time step of 0.01 s, and the displacements of phase 1 subtracted as the manual's reset of displacements does.

The building vibrates with a period of 0.904 s in Tellus 2D against 0.948 s in the tutorial, a dominant frequency of 1.10 Hz against about 1 Hz, and its first swing from the release to the first trough is 10.6 mm against 10.8 mm (Figure 9). The amplitude decays at a damping ratio of 0.040 against 0.058 from the successive peaks of the tutorial figure. The two programs differ, however, in the displacement at the release, 7.31 mm against 5.75 mm, and Tellus 2D vibrates about a mean of 0.552 mm after 2 s, so its oscillation is shifted towards the side of the load, whereas the tutorial extremes are almost symmetric about zero. A run with both sands linear elastic at their small-strain stiffness at mid-depth, $E_0 = 2G_0^{\text{ref}}(1+\nu)(\sigma'_3/p_{\text{ref}})^m$ with σ'_3 of about 60 and 180 kPa, starts from 5.57 mm, within 3% of the tutorial, and vibrates at 1.15 Hz about a mean of -0.516 mm. The static push of phase 2 is therefore stiffer in the tutorial, close to the small-strain stiffness, than in the HS small of Tellus 2D, which develops irreversible strains under

the building's weight and the horizontal load and does not recover them when the load is released. The difference lies in the static response of HS small near the small-strain range rather than in the dynamic solution, and the published figure does not allow us to locate it further.

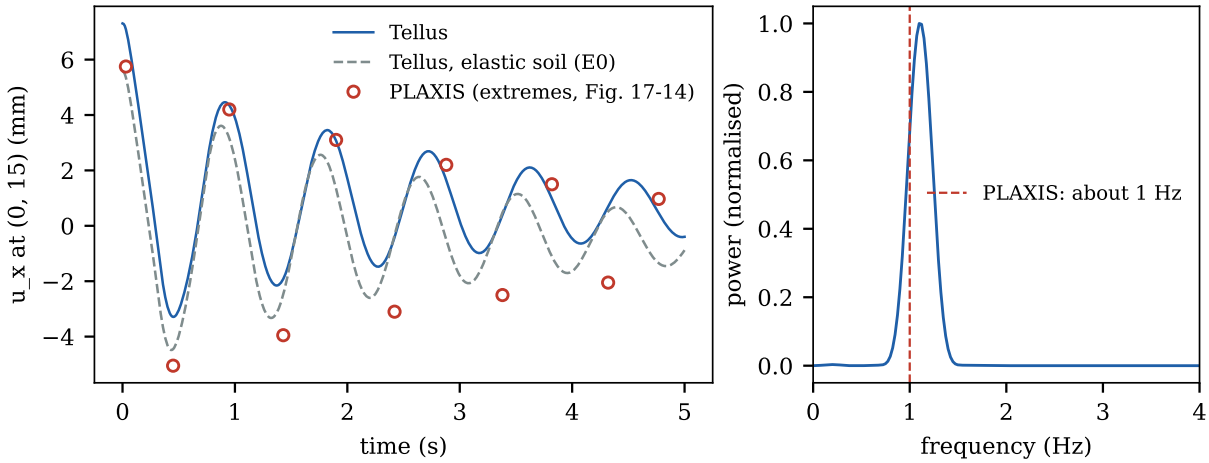


Figure 9: PLAXIS 2D tutorial lesson 17, free vibration after release: (a) horizontal displacement of the top of the building, Tellus 2D with HS small and with elastic soil, and the extremes read from Figure 17-14 of the manual, (b) power spectrum of the Tellus 2D displacement.

5.3 Nonlinear response of Bangkok clay: Motus

The Bangkok profile of the Motus examples has 14 m of soft to medium clay (V_s = 70–120 m/s, s_u = 13–30 kPa) over stiff clay and sand to 600 m, on rock of 2,000 m/s, and it was shaken by a stochastic M7 motion at 150 km with a PGA of 0.075 g as an outcrop motion. Motus solves it as a nonlinear shear beam with the GQ/H backbone fitted to the strength of each clay layer, and as an equivalent linear profile. Tellus 2D modelled the clays as elastic-perfectly plastic Tresca soils with $G = \rho V_s^2$ and the given s_u, the sands as elastic, and all layers with Rayleigh damping of 2% at 0.3 and 10 Hz, so it differs from Motus in its stress-strain curve before failure and in its damping. Tellus 2D gives a surface PGA of 0.260 g, against 0.107 g for the nonlinear and 0.156 g for the equivalent linear solution of Motus, and a spectral peak of 0.748 g at 0.653 s against 0.447 g at 1.31 s and 0.739 g at 2.09 s (Figure 10). The clay of Tellus 2D does yield, since the mean shear strain across the upper soft clay reaches 0.809%, five times its yield strain s_u/G = 0.163%, and 0.686% remains at the end, but it stays at $G = \rho V_s^2$ and 2% damping until it yields, whereas the GQ/H clay of Motus softens and dissipates energy from the smallest strains and reaches 7.94% in the same layer. Tellus 2D therefore transmits and amplifies the short periods that Motus filters out, and the damped Newmark rule (β = 0.3025, γ = 0.6) changes its PGA only to 0.252 g, so the difference is not numerical. The comparison shows how much the result depends on the soil model in a soft clay that reaches its strength, and the paper does not claim agreement here: an elastic-perfectly plastic clay is not suitable for a site response analysis of soft clay, for which HS small or a hyperbolic model with damping should be used.

6 Comparison with a downhole array

The Garner Valley Downhole Array (GVDA) in southern California records the same earthquakes at the surface and at depths of 6, 15, 22, 50 and 150 m in 22 m of alluvium over decomposed granite, and its shear wave velocity was measured with a suspension log at 1 m spacing [26]. Six earthquakes of magnitude 3.6 to 7.1 at epicentral distances of 6.5 to 248 km, recorded between 2013 and 2019 with surface PGA of 0.007 to 0.08 g, give 12 horizontal records, each corrected for the orientation of the borehole sensors. Tellus 2D modelled the 50 m profile of 51 layers as a tied column on a rigid base driven by the recorded 50 m motion as a within motion, linear elastic with Rayleigh damping of 1, 2 or 4% at 1 and 15 Hz, since the strains of these events are small.

At 2% damping the mean log ratio of the computed to the recorded surface spectra over 0.05–1 s is 0.326 with a standard deviation of 0.119 between records, it falls to 0.177 at 4% damping, and the surface PGA is on average 1.32 times the recorded one (Figure 11). The over-prediction is largest at 0.1–0.5 s, and it is the behaviour reported for one-dimensional analyses at many vertical arrays, where scattering by lateral variations of the soil and damping larger than that measured in the laboratory reduce the short-period motion at the surface [27]. The residuals peak at

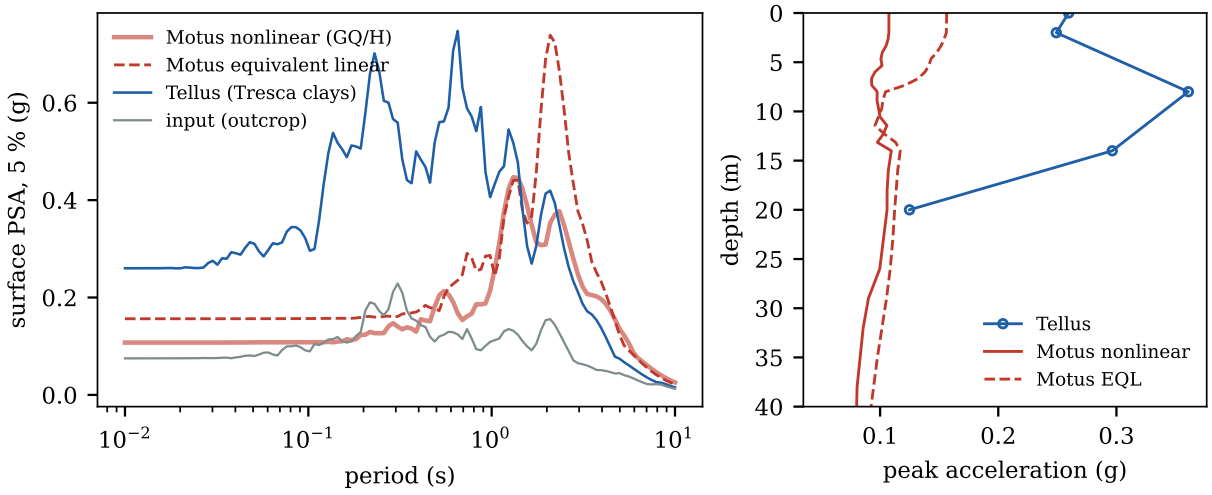


Figure 10: Bangkok soft clay profile under a stochastic M7 motion of 0.075 g: (a) surface spectra of Tellus 2D (Tresca clays) and of Motus (nonlinear GQ/H and equivalent linear), (b) peak acceleration in the top 40 m.

the profile's resonances near 0.22 and 0.45 s, where the model concentrates its response more sharply than the site does, and matching the 2% Rayleigh damping at the site frequency f_0 = 1.64 Hz and $5f_0$ instead of at 1 and 15 Hz lowers the mean log ratio to 0.234 ± 0.108. The array therefore confirms the spectral shape and the depth distribution of the peak accelerations of the analysis, while its level depends on the damping, which a one-dimensional model has to calibrate to the site.

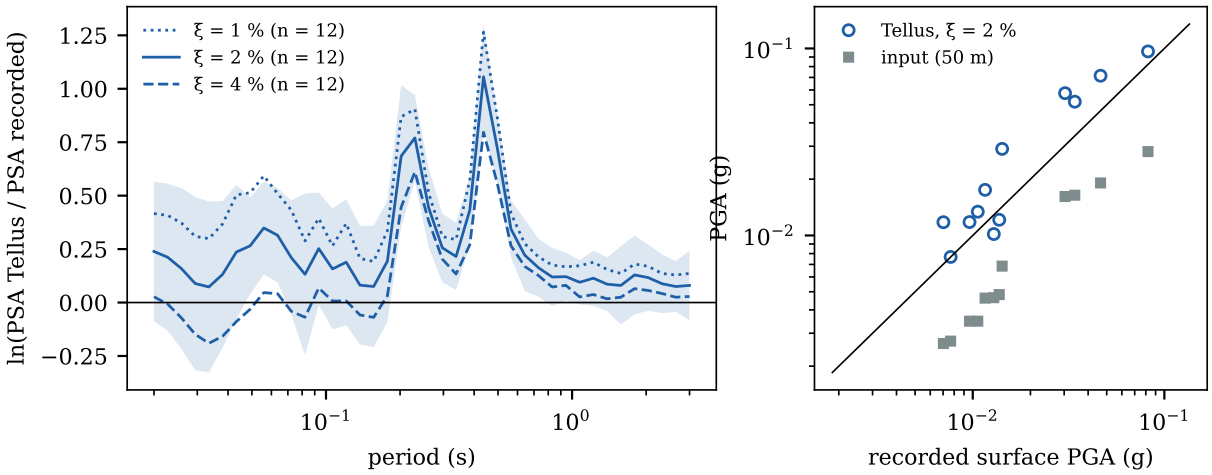


Figure 11: Garner Valley Downhole Array, 12 records driven by the recorded 50 m motion: (a) mean log ratio of the computed to the recorded surface spectra for three damping ratios, with ±1 standard deviation at 2%, (b) computed and recorded surface PGA.

7 A model of cyclic pore pressure: PM4Silt

The soil models of the static analysis do not generate pore pressure under cyclic shearing, so Tellus 2D was given PM4Silt [9], the effective-stress model for low-plasticity silts and clays that Boulanger [7] used for Austrian Dam and one of the models most used in practice for the seismic analysis of embankments on such soils. PM4Silt is a stress-ratio controlled, critical-state compatible, bounding surface plasticity model in the framework of Dafalias and Manzari [13] and PM4Sand [8], with a small yield cone around a back-stress ratio, bounding and dilatancy surfaces that depend on the state parameter, and a fabric tensor that lets an undrained element accumulate shear strain cycle by cycle once its pore pressure is high. It has three required parameters, the critical-state undrained strength $s_{u,cs}$ (or its ratio to σ'_{vc}), the shear modulus coefficient G_o and the contraction rate h_{po}, and twenty secondary parameters with default values. Tellus 2D implements it from the equations of the report (its Table 2.1 and Sections 2–4), not from

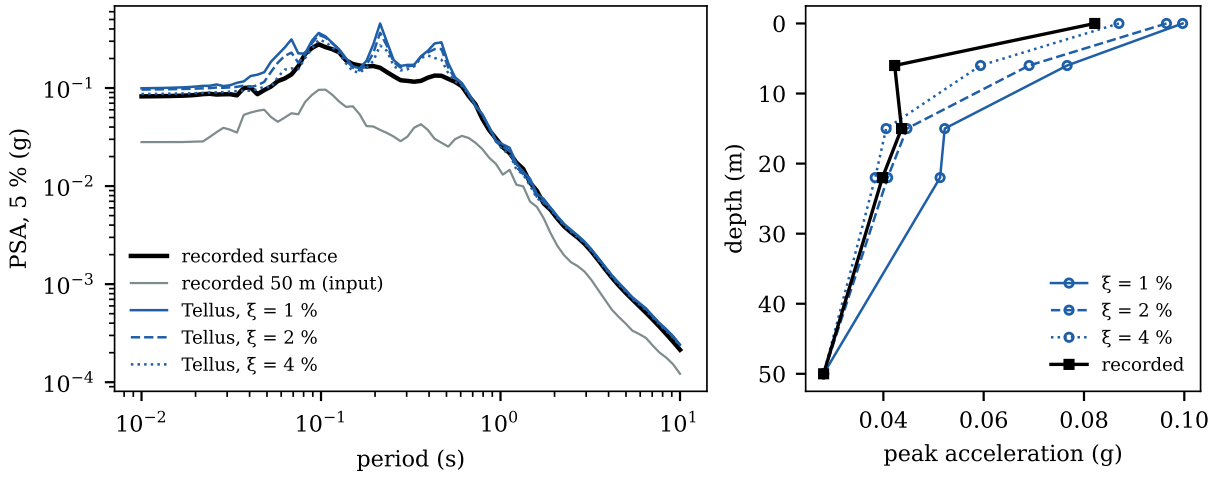


Figure 12: GVDA, 2016 M_w 5.2 Borrego Springs earthquake, east component: (a) 5%-damped surface spectra recorded and computed, (b) peak acceleration with depth.

any existing code, in the plane-strain form of the report with the in-plane mean stress $p = (\sigma'_{xx} + \sigma'_{yy})/2$. As in the procedure of Boulanger [7], a PM4Silt soil is Mohr-Coulomb in static phases and becomes PM4Silt at the start of its first dynamic phase, where the model's FirstCall positions the critical state line for $s_{u,cs}$ and sets the back-stress ratio to the current stress ratio. The strength $s_{u,cs}$ can be given, taken as a ratio of σ'_{vc}, or computed at each stress point from its consolidation stresses by the two-stage procedure of Duncan et al. [14], which Boulanger [7] used for the dam. Undrained (A) soil then develops pore pressures from the bulk stiffness of the water, which corresponds to the coupled analysis without flow of the FLAC runs.

The stresses are integrated explicitly in sub-steps with a return of the stress ratio to the yield cone after each, and the size of the sub-steps decides whether the model works at all. The yield cone has a radius of $m/\sqrt{2} = 0.007$ in stress ratio, so sub-steps of a fixed strain of 2×10^{-5} moved the stress ratio by nearly a cone radius each, the loading direction chattered and a monotonic simple shear test stalled at 22% below $s_{u,cs}$. Tellus 2D therefore sizes the sub-steps so that the elastic change of the stress ratio in one of them is at most a quarter of the cone radius, after which the monotonic test reaches $s_{u,cs}$ exactly. The global iterations use the initial stiffness with Anderson acceleration, and the iteration matrix takes a modulus at which both its shear and its bulk stiffness, with the soil's Poisson's ratio, are at least those of PM4Silt at the stress, because a bulk stiffness below the model's left the iterations oscillating.

Five points that the report leaves open or states twice had to be settled, and each is stated here because it changes the results. The dense-of-critical bounding surface uses $M^{b,max} = 2\sin\phi_{max}$, for which the report gives no value, so Tellus 2D takes $\phi_{max} = \phi_{cv} + 20°$ (at most 70°) as a parameter, although values from 61° to 80° changed the cycles to 3% strain of the Austrian Dam calibration by less than one cycle. The apparent initial back-stress ratio $\boldsymbol{\alpha}_{in}^{app}$ is described per component as the minimum value it has ever had, but no smaller than zero, which pins it at zero in symmetric cycles, whereas the report also requires $(\boldsymbol{\alpha} - \boldsymbol{\alpha}_{in}^{app}) : \mathbf{n} = 0$ at the start of unloading, so Tellus 2D takes it as the start of the current loading branch, that is the last reversal, or the one that a small unload-reload loop interrupted once the loading passes the previous reversal point. This choice matters more than any other, because the first reading reduced the contraction after each reversal so much that the published cyclic strengths needed a quarter of the published h_{po}, whereas the second reproduces them with the published h_{po}. The plastic modulus with the reversal factor C_{rev} is evaluated in the equivalent form $Gh_o\sqrt{b}/[(\boldsymbol{\alpha} - \boldsymbol{\alpha}_{in}^{true}) : \mathbf{n}\,\varphi(x) + C_{\gamma 1}]$ with $x = (\boldsymbol{\alpha} - \boldsymbol{\alpha}_{in}^{app}) : \mathbf{n}$ and $\varphi(x) = (e^x - 1)/x$, which equals the report's expression and stays finite where its denominator would vanish. The factor C_{pmin} is defined as a linear ramp from $2p_{min}$ to $8p_{min}$ in the text and as $1/[1 + (p_{min}/p)^2]$ in Table 2.1, so the ramp, whose role in limiting the excess pore pressure the text explains, is used in contraction and the Table's form in the dilatancy factor A_d. Lastly, a state beyond the bounding surface softens with a negative plastic modulus, which a dense soil needs to come down to the critical state but which made the implicit iterations fail where loose initial stresses lay beyond the surface, so loose of critical the plastic modulus is held at zero there, which with $n^{b,wet} = 1$, as in the Austrian Dam calibrations, removes no softening that the model would otherwise show in undrained shear.

In undrained monotonic simple shear the implementation reaches the critical-state strength exactly, 25.0, 50.0 and 75.0 kPa at 30% strain for $s_{u,cs}/\sigma'_{vc} = 0.25$, 0.5 and 0.75 with the example calibrations of the report, and the loose specimen peaks at 27.3 kPa before it softens to $s_{u,cs}$, as in the report. With the published parameters, including h_{po}, the cyclic strength curves follow those of FLAC with somewhat fewer cycles to a given strain (Figure 13a,b). The specimen with $s_{u,cs}/\sigma'_{vc} = 0.5$ needs 0.5, 2.5, 5.5, 11.5 and 27.5 cycles to reach 3% strain at CSR 0.48 to 0.275

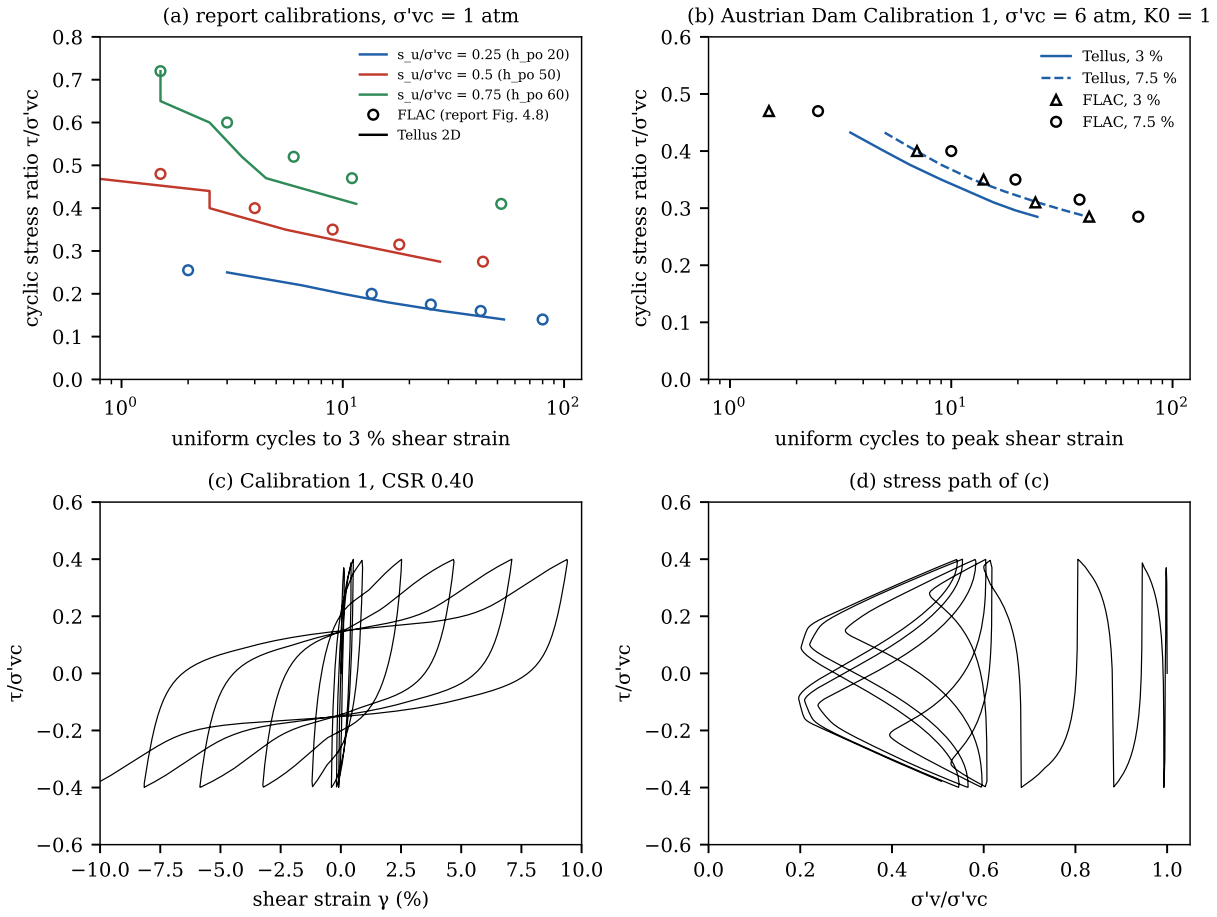


Figure 13: PM4Silt in undrained cyclic direct simple shear with the published parameters: (a) cycles to 3% shear strain for the example calibrations of the report at 1 atm and $K_0 = 0.5$ [9], (b) Austrian Dam Calibration 1 ($h_{po} = 40$) at 6 atm and $K_0 = 1$ against the FLAC results of Boulanger [7], (c) and (d) stress-strain response and stress path of Calibration 1 at CSR 0.40.

against 1.5, 4, 9, 18 and 43 in the report, the looser specimen 10 and 28 cycles at CSR 0.20 and 0.16 against 13.5 and 42, and the denser one 2.5, 3.5 and 4.5 cycles at CSR 0.60, 0.52 and 0.47 against 3, 6 and 11 but only 11.5 against 52 at CSR 0.41, the largest difference, which lies where the dense specimen dilates strongly and its response depends most on the fabric terms. The Austrian Dam calibrations of Boulanger [7] with their own h_{po} of 40 and 80 behave in the same way. Calibration 1, consolidated isotropically as its strength $s_{u,cs} = 0.47\,\sigma'_{vc}$ at 6 atm implies, reaches 3% strain in 5.0, 9.0 and 15.6 cycles at CSR 0.40, 0.35 and 0.31 against 7, 14 and 24 in FLAC, so its cyclic stress ratio for 3% in 30 cycles is 0.28 against their 0.30, and Calibration 2 reaches it in 3.5 and 12.5 cycles at $\tau_{cyc}/s_{u,cs} =$ 0.85 and 0.66 against about 7 and 24. Consolidated with $K_0 = 0.5$ instead, Calibration 1 reaches 3% strain in half the cycles, because the initial anisotropic stress ratio brings the element closer to its bounding surface, so the consolidation state of the element tests, which Boulanger [7] did not state, matters as much as the remaining difference between the implementations. The stress paths look alike, because the vertical effective stress falls to about 0.6 of its initial value within a few cycles and the element then cycles through phase transformation with shear strains growing by a few per cent a cycle down to $\sigma'_v/\sigma'_{vc} = 0.20$, against 0.2–0.25 in their Figure 13 (Figure 13c,d). The remaining differences, a factor of about 1.5 in cycles and up to 4.5 for the densest specimen, come from details of the FLAC implementation that the report does not give, so a user of Tellus 2D should still check a calibration on element tests in Tellus 2D and adjust h_{po} if the target cycles are not met, as the calibration procedure of the report prescribes.

8 Case study: Austrian Dam in the 1989 Loma Prieta earthquake

Austrian Dam is a 55 m high compacted earthfill dam in Santa Clara County, California, completed in 1951, and it was the earth dam most heavily damaged by the M_w 6.9 Loma Prieta earthquake of 1989 [15]. Its crest settled 750 to 859 mm over the central 140 m of its 230 m length, longitudinal cracks up to 300 mm wide and 4.3 m deep opened in the upper parts of both faces, the downstream face bulged near its toe, and the upper part of the embankment had to be

removed and rebuilt. The fill is a nearly homogeneous low-plasticity clayey sand and gravel (fines 16–44%, PI 11–15), whose effective friction angle from isotropically consolidated undrained triaxial tests is 41–43° with no cohesion, and whose small-strain shear modulus from resonant column tests is $G_{\max} = 2280\, P_a (p'/P_a)^{0.6}$ [7]. The reservoir stood about 30 m below the crest after three years of drought, and piezometers read one day before and two days after the earthquake recorded excess heads of 2.7 to 16.8 m in the saturated lower part of the fill, which dissipated slowly over the following weeks. Boulanger [7] analysed the dam with PM4Silt in FLAC, calibrated to these tests, and obtained crest settlements of 610 to 1060 mm with four recorded input motions, so the case offers both a measured failure and a calibrated analysis with an advanced model.

The Tellus 2D model uses the maximum section of Boulanger [7], digitised from their Figure 2 at its true scale (the traced faces have slopes of 2.3–2.6, 3.0 and 3.4–3.5 against the labelled 2.5, 3 and 3.5), with the fill above the measured phreatic line drained and the fill below it undrained, on 25 to 35 m of rock (V_s = 1000 m/s) with a rigid base and free-field sides (Figure 14). The initial stresses come from gravity loading with the reservoir at El. 312.1 m, after which the dynamic phase applies 25 s of a horizontal record as the within motion at the base. The records that Boulanger [7] used at Corralitos and at the abutment of Lenihan Dam are available only from archives that need a registered login, so this paper used the Gilroy Gavilan College rock record of the same earthquake (GIL067, 0.357 g, about 28 km from the epicentre), at its own amplitude and scaled to 0.478 g, the peak of the Corralitos channel 1 record, and without the vertical component, whose omission reduced the settlements of Boulanger [7] by 5–12%. The fill was first modelled with the soil models of the static analysis: Hardening Soil with its unloading-reloading modulus at half of $G_{\max}$ and Rayleigh damping of 5% at 2 and 10 Hz, Mohr-Coulomb with the modulus of $G_{\max}$ at 250 kPa, and a total-stress model of current practice in which the saturated fill is a Tresca soil with $s_u = 0.4\,\sigma'_{vc}$, the critical-state strength ratio of Boulanger [7], or with the cyclic strength $0.22\,\sigma'_{vc}$, which is 0.55 times that, the ratio of cyclic to monotonic strength in the cyclic triaxial tests on the dam's fill.

The saturated fill was then modelled with PM4Silt (Section 7), with all the parameters of the two calibrations of Boulanger [7] (G_o = 2280, n_G = 0.6, h_{po} = 40 and 80, ϕ_{cv} = 41°, e_o = 0.394, $n^{b,wet}$ = 1.0, z_{max} = 5, C_z = 50, C_{GD} = 6), $s_{u,cs}$ computed at each stress point from its consolidation stresses by their two-stage envelopes (Calibration 1: d_R = 0.37 atm and ψ_R = 22.4° for K_c = 1, ϕ' = 43.4° for $K_c = K_f$, Calibration 2: 0.54 atm, 28.6° and 41.1°) and limited to σ'_{vc}, and Rayleigh damping of 0.5% at 1 Hz as in their analyses. Three choices differ from their procedure, and each was needed to start the shaking from a stable dam. The gravity loading uses $\nu = 1/3$, so that K_0 = 0.5 as in their initial stresses, because $\nu = 0.2$ left $\sigma'_h/\sigma'_v = 0.25$ and most saturated stress points beyond the bounding surface. The static shear $q/2$ of the Mohr-Coulomb stresses still exceeded the two-stage $s_{u,cs}$ at 1,755 of the 12,468 saturated stress points with Calibration 1, so $s_{u,cs}$ was raised to $1.05\,q/2$ where it was smaller, which changed it at 2,420 points (19%) with Calibration 1 and 899 points (7%) with Calibration 2. Lastly, the unsaturated fill stayed Mohr-Coulomb with ϕ' = 41°. With PM4Silt in the whole fill, as in FLAC, the drained fill near the faces lies loose of critical with a bounding friction angle of about 26°, and every analysis stopped in the strongest shaking at t = 2.8–3.5 s, because the implicit iterations found no equilibrium where this fill began to flow, also with the strength raised to keep every stress point inside the bounding surface and with steps accepted out of balance, whereas the explicit scheme of FLAC carries such zones through.

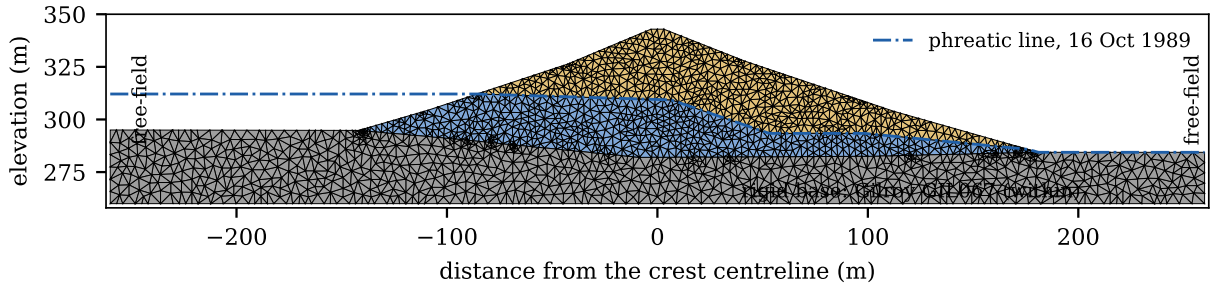


Figure 14: Austrian Dam model: rock (grey), unsaturated fill (yellow) and saturated fill (blue, in four bands of undrained strength for the total-stress model), the phreatic line measured one day before the earthquake, and the mesh of 3,731 15-node elements.

The crest settlement is therefore not used to judge the analyses. The settlement of the dam came from the slumping of both faces, in which the unsaturated fill takes part, so a model that keeps that fill stiff and strong cannot be expected to reproduce it, and the substitute input motion adds an uncertainty of its own, because the Gilroy record is weaker than the Corralitos record at periods above about 0.3 s even when scaled to the same peak. The comparison rests instead on the response that the saturated fill controls, the accelerations it transmits to the crest and the excess pore pressures it develops (Table 2).

PM4Silt brings the crest accelerations down to those of the FLAC analyses, which is the clearest effect of cyclic

Table 2: Austrian Dam: peak crest acceleration and the excess heads at the piezometers at the end of 25 s of shaking, against the excess heads read two days after the earthquake and the FLAC-PM4Silt analyses of Boulanger [7]. The heads are read in the element nearest each sensing point, taken from their section drawing.

Model of the saturated fill	Input PGA (g)	Crest PGA (g)	P-1 (m)	P-2 (m)	P-4 (m)	P-6 (m)
Hardening Soil, 5%	0.357	1.10	9.2	3.2	4.2	4.4
Hardening Soil, 5%	0.478	1.31	9.3	2.8	2.7	5.9
Hardening Soil, 2%	0.478	1.77	9.5	3.9	1.4	5.5
Mohr-Coulomb, 5%	0.357	0.97	11.9	4.0	8.1	6.9
Mohr-Coulomb, 5%	0.478	1.15	12.2	4.4	6.8	8.8
Mohr-Coulomb, 2%	0.478	1.23	12.5	4.6	8.4	8.6
Total stress, $s_u = 0.22\,\sigma'_{vc}$	0.357	1.03	–	–	–	–
Total stress, $s_u = 0.22\,\sigma'_{vc}$	0.478	1.12	–	–	–	–
Total stress, $s_u = 0.4\,\sigma'_{vc}$	0.478	1.41	–	–	–	–
PM4Silt, Calibration 1	0.357	0.59	9.6	6.7	14.8	9.9
PM4Silt, Calibration 1	0.478	0.65	11.3	7.8	15.8	9.7
PM4Silt, Calibration 2	0.357	0.58	9.4	6.0	13.8	10.2
PM4Silt, Calibration 2	0.478	0.72	9.8	6.5	15.6	9.6
Measured	–	–	16.8	2.7	4.0	15.2
FLAC-PM4Silt	0.41–0.63	about 0.7	10.0–19.6	6.6–11.3	6.3–12.2	2.5–6.9

softening that the analyses show. The models without a cyclic pore pressure model transmit peak crest accelerations of 0.97–1.77 g, about twice the 0.7 g of FLAC, whereas PM4Silt gives 0.58–0.72 g with the two calibrations and both input levels, because the saturated fill softens as its pore pressure rises and limits what it can pass upward, as in FLAC.

The excess pore pressures are of the measured magnitude, but they do not single out PM4Silt, and they err in the same places as those of FLAC (Figure 15). With Calibration 1 and the input at 0.48 g the excess heads are 11.3 m at P-1, 7.8 m at P-2, 15.8 m at P-4 and 9.7 m at P-6, against the measured 16.8, 2.7, 4.0 and 15.2 m, so P-1 and P-6 are under-predicted and P-2 and P-4 over-predicted, as in the FLAC analyses, although P-6 is closer to the measurement than in FLAC. The largest excess pore pressures, up to 220 kPa, develop near the base of the fill just downstream of the centreline, where FLAC also had its largest values of about 300 kPa. The Hardening Soil and Mohr-Coulomb fills, which are undrained (A) but do not contract under cyclic shearing, also end with excess heads of 9–12 m at P-1 and come closer to the small heads measured at P-2 and P-4, because the pore pressure of an undrained soil also follows the changes of total mean stress as the dam deforms, so these heads, read at a few points close to the phreatic line where they change rapidly, cannot by themselves distinguish the models.

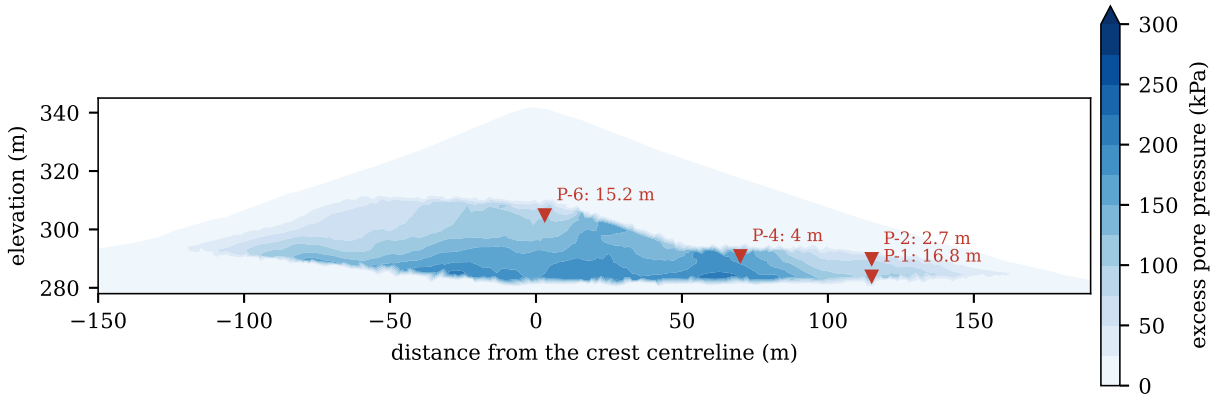


Figure 15: Austrian Dam, PM4Silt Calibration 1 with the input at 0.48 g: excess pore pressure at the end of shaking, with the piezometers and their measured excess heads [7].

The case therefore shows what the PM4Silt of Tellus 2D can and cannot yet do for an earth dam. It generates pore pressures in the saturated fill of the measured magnitude and with the distribution of the FLAC analyses, and it reduces the crest accelerations to theirs, which the models without cyclic pore pressure do not, but the implicit solution cannot yet carry a loose unsaturated fill through the flow that strong shaking starts in it, so a comparison of crest settlement with the measurement and with FLAC needs either PM4Silt throughout the fill with a solution that follows such flow, or the records of Boulanger [7].

9 Case study: the Bangkok basin in the 2025 M_w 7.7 Myanmar earthquake

The M_w 7.7 earthquake on the Sagaing fault on 28 March 2025 shook Bangkok, about 600 km from the epicentre, more strongly than any earthquake recorded there before, with peak ground accelerations of about 0.02 g and peak velocities of about 10 cm/s at the strong-motion stations of the Thai Meteorological Department (TMD) in the city, and with an amplification at long periods that Ornthammarath et al. [10] attribute to the thick soft sediments of the Bangkok basin. The case tests whether the dynamic analysis of Tellus 2D can carry a section of a real sedimentary basin tens of kilometres long through a long, distant earthquake record, and what such a section adds to the one-dimensional analyses of current practice.

9.1 Model

The section runs 54 km from west to east across Bangkok at about latitude 13°45′N through six of the twenty sites at which Poovarodom and Jirasakjamroonsri [12] measured shear-wave velocity profiles down to the bedrock with microtremor arrays (Figure 16). Their positions were read from the site map of that study and projected on the line, and each site gives its average shear-wave velocities to 30, 100 and 500 m and the depth of the velocity contrast that marks the bedrock. The bedrock lies at 636 m below the western site BKK05, deepens to 727–737 m under central Bangkok, rises to 410 m at BKK12 and lies at 650 and 634 m at the eastern sites, and it is taken as linear between the sites. Between the midpoints of neighbouring sites each zone takes the interval velocities of its site, 114–162 m/s in the upper 30 m, 285–509 m/s from 30 to 100 m and 564–899 m/s from 100 m to 500 m or to the bedrock where it is shallower, with about 1000 m/s from 500 m to the bedrock, as Poovarodom and Jirasakjamroonsri [11] found, and 2000 m/s in the bedrock. The upper 30 m, which comprise the weathered crust, the soft Bangkok clay of 15–20 m and the first stiff clay, are one layer with the travel time of the measured profile, which represents them for periods above about 0.5 s, so elements of half a wavelength at 1 Hz, limited to the thickness of each layer, give 34,903 fifteen-node triangles and 283,702 nodes. The ground surface is flat, because Bangkok lies 0.5–2 m above sea level, the sides are free-field columns, the base at 850 m is compliant with the bedrock below it, and all layers are elastic with Rayleigh damping of 2% at 0.2 and 2 Hz, since the motion of about 0.02 g leaves the soft clay close to its small-strain stiffness.

The input had to come from a rock site outside the basin, and only one public record of the event was usable. The broadband sensors of the TMD network that are open through the IRIS data centre clipped at the rock sites near Bangkok, the Bangkok station of that network returned no signal, and the strong-motion records of the TMD stations in the city are not public. The strong-motion accelerometer of the Global Seismographic Network station IU.CHTO at Chiang Mai, on rock 473 km from the epicentre, recorded the event without clipping, with a peak of 0.0115 g on its first horizontal component, so 300 s of that component, filtered between 0.05 and 20 Hz, was applied as the outcrop motion at the compliant base, with a time step of 0.02 s. Chiang Mai lies closer to the rupture than Bangkok, so this input is likely stronger than the rock motion beneath the city.

9.2 Results

The section amplifies the input by a factor of 2.4–2.9 in peak acceleration at the sites, and by 4.5–7.7 in spectral acceleration at periods of 0.6–1.5 s, the period range of the soft clay and the upper sediments (Figure 17). The surface peak acceleration rises from about 0.027 g in the west to 0.033–0.039 g in the east, and the largest spectral amplification, 7.7 at 1.0 s, lies above the bedrock high at BKK12, where the stiffer sediments under the soft layer end at a shallower bedrock. A second band of amplification at about 3 s appears between BKK13 and BKK16 in the east, which agrees with the zonation of Poovarodom and Jirasakjamroonsri [11], who found high spectral accelerations at about 2 s only in that part of the city. The strong shaking at the surface lasts from about 95 to 125 s of the record, as in the input, with long-period oscillations that continue for a few tens of seconds longer.

One-dimensional columns with the same profiles at the six sites give almost the same response, so along this section the basin acts as a set of one-dimensional columns for vertically incident waves (Figure 18). The two-dimensional spectra lie within 0.90–1.08 of the one-dimensional ones at periods of 0.7–2.5 s at every site, and the peak accelerations within 8%, with a larger difference only at BKK13 at periods of 3–4 s, where the bedrock dips from the high at BKK12. The relief of the bedrock, about 300 m over 14 km, is gentle compared with wavelengths of several kilometres at these periods, so it neither focuses nor traps the vertically incident waves, and the largest amplification above the bedrock high comes from the profile of that site rather than from its geometry. This is a useful result for practice, because it shows that one-dimensional analyses with site-specific profiles suffice for the response to vertically incident waves at these sites, whereas the two-dimensional section remains necessary where the bedrock or the soft clay changes over distances of a few wavelengths, such as the edges of the basin.

The computed motions agree with the records of the event in velocity and in the period of amplification but exceed them in peak acceleration. The peak velocities at the sites, 10–14 cm/s, agree with the about 10 cm/s recorded in

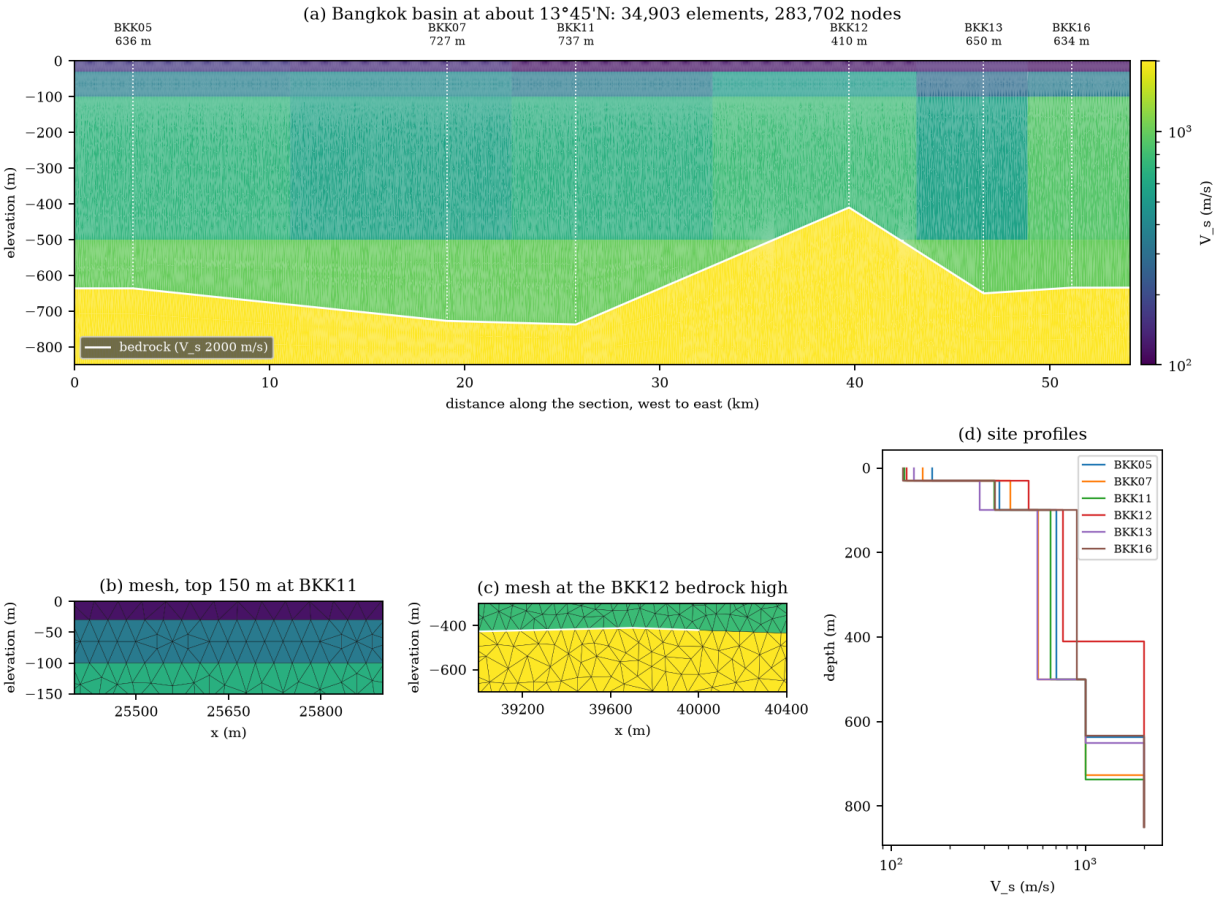


Figure 16: Bangkok basin section at about 13°45′N: (a) shear-wave velocity along the section from the microtremor sites of Poovarodom and Jirasakjamroonsri [12], with their bedrock depths, (b) and (c) the mesh near the surface and at the bedrock high below BKK12, (d) the site profiles.

Bangkok, and the amplification of 2.4–2.9 in peak acceleration and the strongest spectral amplification near 1 s agree with the long-period amplification inside the basin that Ornthammarath et al. [10] report, but the peak accelerations of 0.027–0.034 g are 1.3–1.7 times the about 0.02 g recorded. The input is the most likely cause, because the Chiang Mai station lies closer to the rupture than Bangkok, and the same amplification applied to a rock motion of about 0.007 g beneath Bangkok would give the recorded peak, but this cannot be confirmed without a rock record near the city. The comparison also has limits of its own, since it rests on reported peak values rather than on the TMD accelerograms, a vertically incident input cannot produce the surface waves that a distant earthquake brings into a basin at long periods, the model ends at the bedrock depths of the microtremor surveys, and its upper 30 m represent the soft clay only above about 0.5 s. A comparison of time histories and spectra station by station therefore needs the strong-motion records of the TMD stations in Bangkok, which lie within a few kilometres of the section.

The analysis took 5,358 s on the RTX 5080 for 15,000 time steps, sharing the box with a second run, and the six columns 75–134 s each. Tests of a uniform section of the same basin showed how far the analysis scales on one GPU, because a section of 100 km with 1.74 million nodes (3.5 million unknowns) needed 6.2 GB of GPU memory and 19 GB of host memory, whereas the same section with the thin layers of the soft clay kept separate, 5.5 million nodes, filled the 16 GB of the GPU and exhausted the 62 GB of host memory of the box, so layers thinner than the elements needed at the frequency of interest should be lumped as here.

10 Computing cost

All analyses of the verification and of the other programs ran on one rented cloud box with an RTX 5080 GPU and 26 CPU cores, and the Austrian Dam analyses on a second one with 32 cores, where each run of 5000 steps on 30,500 nodes took 75–360 s, and with PM4Silt, whose stress points need up to a few hundred sub-steps each, 570–730 s with four runs at a time. The large two-dimensional models used the cuDSS factorisation on the GPU and the columns the CPU, whose results agree with those of the GPU to 2×10^{-13}. The column problems of the verification took 1–15 s

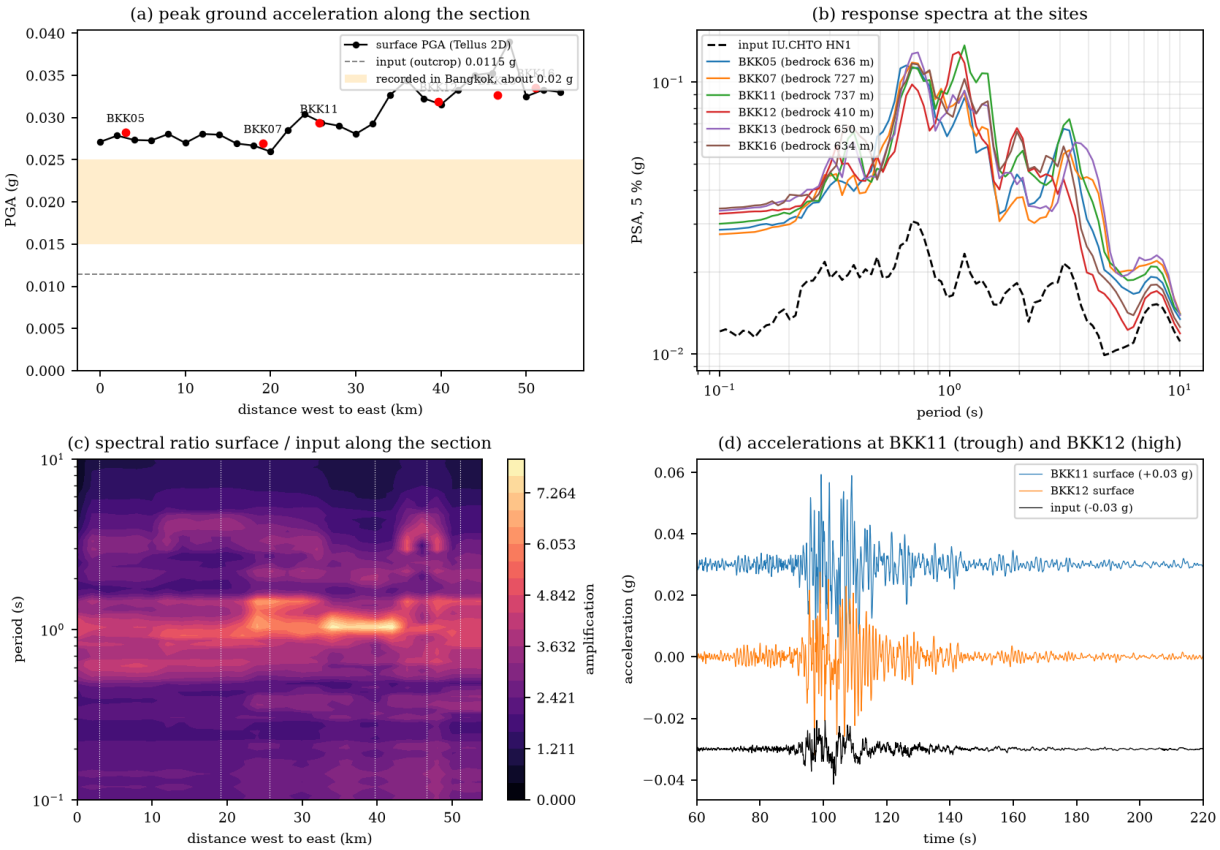


Figure 17: Bangkok basin section in the 2025 Myanmar earthquake with the IU.CHTO record as the outcrop motion: (a) peak ground acceleration along the section, with the input and the level recorded in Bangkok [10], (b) 5%-damped response spectra at the sites, (c) spectral ratio of the surface to the input along the section, (d) accelerations at BKK11 and BKK12.

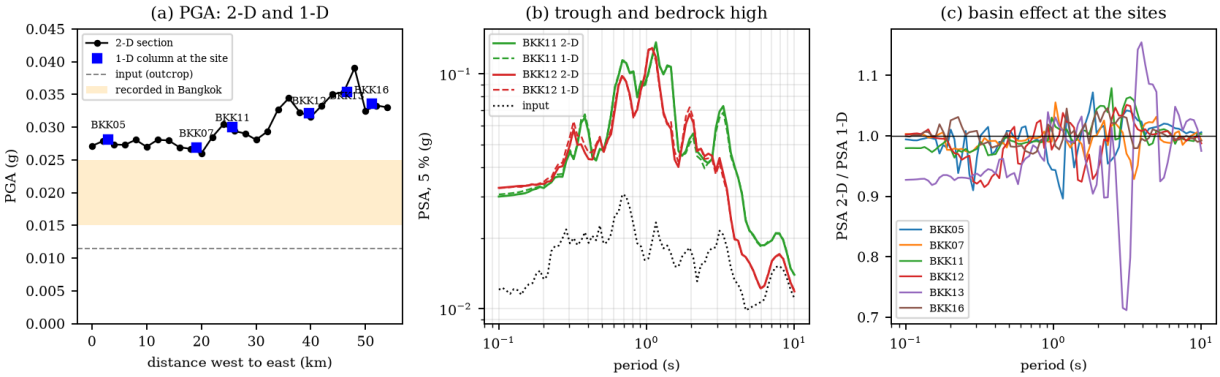


Figure 18: Bangkok basin section against one-dimensional columns with the same profiles at the sites: (a) peak ground acceleration, (b) response spectra at the deepest site (BKK11) and above the bedrock high (BKK12), (c) ratio of the two-dimensional to the one-dimensional spectra at the six sites.

each, the 120 s GVDA records of 24,000 steps about 100 s each, the Strata records 40–100 s, the SHAKE91 problem 28 s, the 600 m Bangkok column of 17,100 steps 87 s, and the tutorial building, with 2,070 elements, 17,030 nodes and 500 steps after its two static phases, 37 s. The elastoplastic sliding block was the slowest problem for its size, 131 s for 1.5 s of motion, because its yielded layer needed many iterations and time steps of half the size. Ten analyses ran at a time, two threads each, and running more at once was slower in total, because the box allowed only about 26 cores to its container. The same problems ran 4–10 times slower on the author's laptop (an Intel Mac), so the dynamic analysis of a real project with long records belongs on such a box, as the final static analyses of the companion paper did [29].

11 Discussion

The verification shows that the time integration, the mass, the Rayleigh damping and the compliant base and free-field boundaries of Tellus 2D are implemented correctly, because they reproduce exact solutions within 3%, and the comparison with DEEPSOIL, Strata and SHAKE91 shows that the two-dimensional program reduced to a column gives the results of established one-dimensional programs. The differences that remain have known causes in the models rather than in the solution. Rayleigh damping is exact at two frequencies only, and the choice of the two frequencies changed the mean spectral difference from SHAKE91 from 2.03% to 5.19% and the bias at the downhole array from 0.234 to 0.326, so an engineer should place them at the site frequency and about five times it, or use HS small for hysteretic damping. Viscous boundaries leave 6–18% of a surface wave reflected in a model 30 m wide and are wrong for an earthquake from the base, for which the free-field or tied sides must be used.

The comparisons have limits that the reader should keep in mind. Most of them are one-dimensional and linear, because these are the problems with exact or well-established answers, so the nonlinear and two-dimensional behaviour rests on fewer checks, which are the sliding block, the tutorial building and the Bangkok profile. The building comparison uses the extremes of a published figure read to about 0.1 mm, and it shows a static deflection 27% larger than in the tutorial that the elastic run traces to the HS small response of the static phase without locating it in the model, and the Bangkok comparison is between different soil models. PM4Silt now generates pore pressures under cyclic loading in low-plasticity silts and clays, but it reaches a given cyclic strain in fewer cycles than in FLAC, it does not cover clean sands, for which PM4Sand would be the analogue, and the implicit integration fails where the initial stresses lie beyond its strength or where a loose unsaturated fill begins to flow in strong shaking, so the static stresses must be checked against $s_{u,cs}$ before shaking, and flow of this kind needs an explicit solution. The free-field columns are elastic, which suits a far field that stays elastic but not one that yields strongly. Two numerical limits appeared in the case study. The dynamic phases of HS small did not converge after gravity loading, because its stiffness switches between G_0 at a strain reversal and G_{ur} in continued loading at thousands of stress points as the iterations change the direction of their strain increments, so the Austrian Dam case was run with Hardening Soil and Mohr-Coulomb, and the dynamic iteration of HS small needs a consistent treatment of these reversals. A dynamic phase that changes nothing static now also starts from the equilibrium of the phase before it, instead of applying the out-of-balance that phase left within its tolerance as a sudden load, which after gravity loading of the dam was 2% of the external forces and stopped the first time step from converging. Dynamic consolidation, which would let the excess pore pressures of PM4Silt dissipate after shaking, a sand model of the same family, axisymmetric models and response spectrum analysis are therefore the next steps.

12 Conclusions

Dynamic analysis in Tellus 2D integrates the equations of motion with the Newmark scheme from the stresses of any static phase, with consistent mass, Rayleigh damping per material and viscous, free-field, tied and compliant boundaries, and an agent can define, run and read it through the program's MCP tools. It reproduces the exact solutions of wave propagation, damping and boundaries within 3%, whereas a viscous side still reflects 13.2–18.4% of a surface wave and the elastoplastic sliding block slides 8.36% more than Newmark's rigid block. Reduced to a column, it agrees with DEEPSOIL, Strata and SHAKE91 within 1–10% in peak ground acceleration and 2–5% in mean spectral acceleration once the Rayleigh damping is placed at the site frequency and about five times it, and it gives the period and first swing of the building of the PLAXIS tutorial within 5%, although its HS small soil makes the static deflection of that building 27% larger than in the tutorial while an elastic soil at the small-strain stiffness gives it within 3%. The remaining differences from other programs come from their models rather than from the solution, namely the frequency dependence of Rayleigh damping and the elastic-perfectly plastic clay against the hyperbolic clay of Motus.

Against measurements the results depend on how well the soil is calibrated, more than on the code. At the Garner Valley downhole array a one-dimensional analysis over-predicts the short-period surface spectra by a mean log ratio of 0.326 at 2% damping and 0.234 with the damping matched at the site frequency, as one-dimensional analyses do at many vertical arrays, so the damping has to be calibrated to the site. PM4Silt, implemented from its published equations, reaches the critical-state strength exactly and the cyclic strengths of the FLAC implementation mostly within a factor of 1.5 in the number of cycles, so a calibration should be checked on element tests in Tellus 2D before it is used. Applied to Austrian Dam in the 1989 Loma Prieta earthquake, PM4Silt in the saturated fill gives excess pore pressures of the measured magnitude and crest accelerations of 0.58–0.72 g, close to the 0.7 g of the published FLAC analysis, which the models without cyclic pore pressure overestimate about twofold, but the crest settlement could not be compared, because the analyses with PM4Silt also in the loose unsaturated fill stopped when that fill began to flow in the strongest shaking. A 54 km section of the Bangkok basin ran through 300 s of the 2025 Myanmar earthquake on one GPU in 90 minutes and reproduced the recorded velocity and the amplification near 1 s, with a peak acceleration

1.3–1.7 times the recorded one that the closer input station may explain, and along this line the basin behaved as one-dimensional columns within 10%.

The program is therefore ready for linear dynamic analyses in plane strain and for effective-stress analyses of silty and clayey fills with PM4Silt calibrated on element tests, provided the boundaries and the damping are chosen as this paper describes, namely free-field or tied sides for an earthquake from the base, viscous sides far from the points of interest, and Rayleigh damping matched at the site frequency and about five times it. Its nonlinear analyses rest on fewer checks, the dynamic phases of HS small do not yet converge after gravity loading, and a loose fill that flows in strong shaking needs an explicit solution, so these, together with dynamic consolidation, a model for clean sands and response spectrum analysis, are the next steps.

Data and code availability

The case inputs, the scripts that run them and make the figures, and the results are in the folder `paper/dynamic` of the Tellus 2D repository, and the tests of Section 4 are in `fem/tests/dynamic.rs`. The reference results of DEEPSOIL, Strata, SHAKE91 and Motus and the GVDA records come from the public sources cited, as collected in the case library of the Motus program.

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